

Customer Segmentation Tool

Frontend & Backend — Complete Explanation

1. What is Customer Segmentation?

Instead of treating all 10,000 customers the same way, a business can divide them into meaningful groups and target each group differently — for example, rewarding VIP customers, or re-engaging users who haven't bought in a while.

2. The RFM Model — The Core Logic

Every customer gets scored on three things:

R — Recency: When did they last buy? (More recent = more valuable)

F — Frequency: How many times do they buy? (More = loyal)

M — Monetary: How much money do they spend?

3. How It Works — Step by Step

Step 1 — Cleaning the Data

Your uploaded file probably has mistakes — empty rows, cancelled orders, or typos. The backend removes all that rubbish automatically before doing any calculation.

Step 2 — Feature Scaling (StandardScaler)

K-Means uses distance math. If one variable (like total spending) ranges from Rs.0 to Rs.1,00,000 and another (like visit count) ranges from 1 to 10, the big number will always dominate. Scaling makes all three RFM variables equal in weight so the AI doesn't get confused by big vs. small numbers.

Step 3 — K-Means Clustering (scikit-learn)

This is where the ML algorithm runs. It groups customers by similarity and repeats the process until the clusters are tight and stable — finding natural segments without being told in advance what the groups should be.

4. Frontend — The Interface (What You See)

Think of the frontend like a restaurant. You sit at the table, look at a menu, order food, and eat. You never go into the kitchen. The frontend is everything the user sees and clicks:

CSV Upload: A simple file upload button where the business analyst drops in their customer data file.

A 'Go' button: You click it, and the analysis begins — no code required.

Results Chart: After a few seconds, a colourful chart appears. Each colour is a customer group (VIPs, bargain hunters, churned users, etc.).

Download option: Save the results for use in reports or marketing campaigns.

That's the entire frontend — simple buttons and charts. Anyone can use it, no technical knowledge needed.

5. Backend — The Brain (What Happens Invisibly)

Think of the backend like the kitchen of the restaurant. The chef works there. You never see it, but all the real work happens there. When you upload your file and click Go, the backend silently performs 5 things:

Step 1 — Cleaning the Data: Removes empty rows, cancelled orders, and typos from your uploaded file automatically.

Step 2 — Calculating 3 Scores Per Customer: For every single customer, it calculates: when did they last buy? How many times did they buy? How much did they spend? These become 3 numbers — R, F, and M.

Step 3 — Balancing the Numbers: One customer might have spent Rs.50,000 while another visited only 3 times. These are on very different scales. The backend equalises them so the AI works correctly.

Step 4 — Running the AI (K-Means): The algorithm groups customers by similarity. This is where the magic happens — it discovers the natural segments on its own.

Step 5 — Sending Results to the Frontend: The backend passes the grouped data back to the interface, which then displays the colourful chart for the user.

6. Key ML Concepts

Unsupervised Learning: You don't tell the algorithm what the groups should be — it discovers them on its own from patterns in the data. There is no 'correct answer' to train on.

K-Means Clustering: An algorithm that assigns each customer to the nearest group centre, then recalculates the centre, and repeats until stable.

StandardScaler: A preprocessing step that makes all features (R, F, M) equally weighted so no single variable dominates.

Simple Summary

The **frontend** is the face — buttons, charts, downloads. Anyone can use it.

The **backend** is the brain — cleaning, calculating, grouping. It does all the work quietly in the background.

You interact with the face. The brain does all the work. You never need to touch it.